Elucidating the Heterogeneity of Wing Morphology in *D. melanogaster* using Wasserstein Geometry

Evan Gibbs[1,4], Ryan Robinett[2], Richard Carthew[3,4], Samantha Riesenfeld[2,4], Shmuel Weinberger[2,4]
York College of Pennsylvania [1] University of Chicago [2] Northwestern University[3]
National Institute for Theory and Mathematics in Biology[4]

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Introduction

- Biological structures and organisms demonstrate remarkable consistency in morphology when comparing organisms within the same species, as well as those closely related. However, to fully understand the constraints underlying this morphological variation, a geometric analysis of biological form is warranted.
- Prior analysis of morphological geometry has historically focused on the selection of a limited number of landmarks residing in given biological structures **[9]**.
- More recently, a landmark-free analysis has emerged using the Pixel-Wise Euclidean Metric for understanding morphological variation in *D. melanogaster* wings **[9]**. Here, wings were conformally mapped to $\mathbb{D}$.

A Computational Morphology Problem

Can we cleverly choose a **metric (geometry)** that novelly quantifies morphological variation, while allowing us to learn more than past methodologies? Notably, we ask this question while using *D. melanogaster* as our model.

Past Approaches: Pixel-Wise Euclidean Geometry

The Pixel-Wise Euclidean Metric

Let $\boldsymbol{X}, \boldsymbol{Y}$ be $\boldsymbol{m} \times \boldsymbol{n}$ grayscale image matrices where

$$X, Y : \{1, \dots, m\} \times \{1, \dots, n\} \to \mathbb{R}$$

We define the **Pixel-Wise Euclidean Distance** to be:

$$d_{m \times n}(X, Y) := \sqrt{\sum_{j=1}^{m} \sum_{k=1}^{n} (X_{jk} - Y_{jk})^2}$$

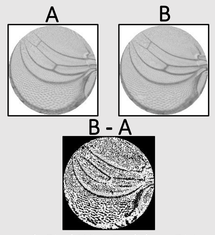


Figure 1: Pixel-Wise Euclidean Distance calculated between two conformally mapped D. melanogaster wings.

- This metric depends on the **upregulation** or **downregulation** of pixels. So, in our toy case, **d(A,B) = d(A,C)**.
- We cannot represent the translation **A → B → C** through a continuous deformation (a single function).
- This also leads to issues of redundancy in Principal Component Analysis (PCA), as well as susceptibility to technical artifacts (such as lighting) **[4]**.

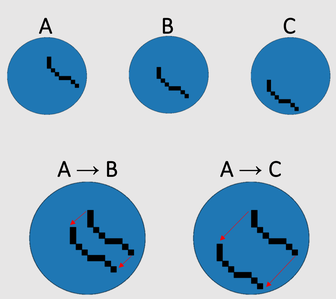


Figure 2: A toy case illustrating the limitation of redundant distance encodings under the Pixel-Wise Euclidean metric.

A New Interpretation of Structure

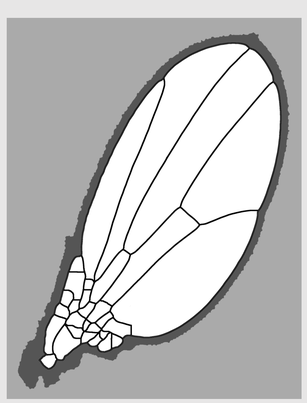


Figure 3: An edge detected and annotated D. melanogaster wing suitable for defining a probability measure.

Let $X = \{1, \dots, m\} \times \{1, \dots, n\}$ denote a wing image embedded in $\mathbb{R}^2$.

Let $S \subseteq X$ be the set of pixels corresponding to the veins or boundaries of the wing.

Define the indicator function $I_S : X \to \{0,1\}$ by

$$I_S(j,k) = \begin{cases} 1, (j,k) \in S \\ 0, (j,k) \notin S \end{cases}$$

This establishes the **support** of a probability measure on $\mathbb{R}^2$.

That is, we can represent all wings as elements embedded in $\mathcal{P}(\mathbb{R}^2)$. **But how can we leverage this new representation of wing structures in $\mathcal{P}(\mathbb{R}^2)$?**

Defining the 2-Wasserstein Metric

The 2-Wasserstein (W_2) Metric via Monge Optimal Transport ☺

Let d_2 be the Euclidean distance on $\mathbb{R}^2$.
Given probability measures μ, ν with well defined PDFs, we define:

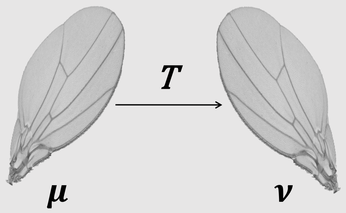
$$S_{\mu\nu} := \{T : \mathbb{R}^2 \to \mathbb{R}^2 \mid T_* \mu = \nu\}$$

The 2-Wasserstein distance between μ and ν is given by:

$$W_2(\mu, \nu) = \sqrt{\inf_{T \in S_{\mu\nu}} \int_{\mathbb{R}^2} d_2(x, T(x))^2 \, d\mu(x)}$$

There are a few more (very) useful variations of this definition. See [1],[2],[3],[5],[8].

- A map $T : \mathbb{R}^2 \to \mathbb{R}^2$ achieving this infimum is known as an **Optimal Transport Map** from μ to ν.
- So, if we can learn T, then this is a novel way to quantify morphological variation, and this method significantly improves upon ⚕.
- Unfortunately, T is **not straightforward** to solve for.



∞-Dimensional Riemannian Manifold on $\mathcal{P}_{ac}(\Omega)$

For a convex and compact domain $\Omega \subset \mathbb{R}^2$, the W_2 metric endows $\mathcal{P}_{ac}(\Omega)$ with a rich geometric structure, especially useful for studying the Optimal Transport Map T. When $\mathcal{P}_{ac}(\Omega)$ is equipped with the W_2 metric, this space admits a formal **Riemannian manifold structure [7]**.

This means that we can meaningfully define notions of

- Distance
- Length
- Angles
- Geodesics⸸
- Exponential and Logarithmic Maps⸸
- Tangent Spaces⸸

Our avenue for exploiting this structure rests heavily upon **[1]**. ⸸ see **Figure 4**

The Unique W_2 Geodesic Path and Realization of Riemannian Logarithms ☼

For $\mu, \nu \in \mathcal{P}_{ac}(\Omega)$, if $T : \Omega \to \Omega$ is the optimal transport map taking μ to ν, then the constant-speed, W_2 geodesic path $\rho : [0,1] \to \mathcal{P}_{ac}(\Omega)$ from μ to ν is given by:

$$\rho_t = ((1-t)Id + tT)_* \mu$$

Consequently, given a Logarithmic map, $\log_u : \boldsymbol{\mathcal{P}_{ac}}(\Omega) \to \boldsymbol{T_\mu \mathcal{P}_{ac}}(\Omega)$, we can realize that the Riemannian Logarithm $\log_u \nu$ of ν at μ, is neatly found as:

$$T - Id$$

The Riemannian Logarithms represent the **tangent vectors** of the tangent space at μ, capturing T and the local structure of the manifold **linearly**.

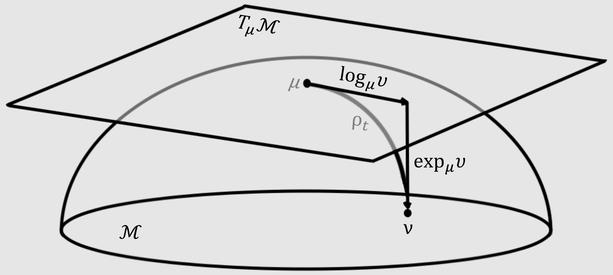


Figure 4: A geometric interpretation of Riemannian geometry on a manifold $\boldsymbol{\mathcal{M}}$. The Riemannian Logarithmic map of ν at μ is denoted by $\boldsymbol{\log_\mu \nu}$, the Riemannian Exponential map by $\boldsymbol{\exp_\mu \nu}$, the Geodesic by $\boldsymbol{\rho_t}$ and the Tangent Space by $\boldsymbol{T_\mu \mathcal{M}}$.

Riemannian Logarithm Visualizations

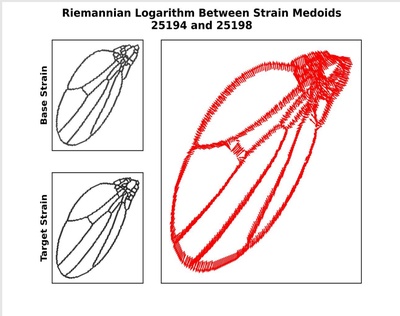


Figure 5: Visual representation of the optimal movement of mass between D. melanogaster Stain Medoids 21594 and 25198.

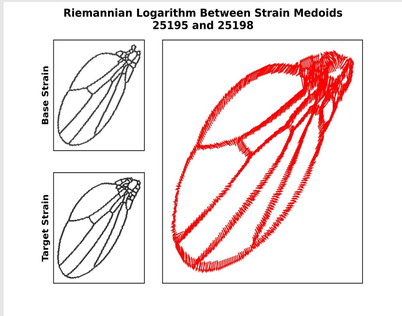


Figure 6: Visual representation of the optimal movement of mass between D. melanogaster Stain Medoids 21595 and 25198.

Scan here to see a continuous deformation of mass between wings! →

Local Gauge of Intrinsic Dimensionality: mSVD

mSVD (multiscale Singular Value Decomposition) is a method used for detecting the intrinsic dimensionality of data **[6]**.

mSVD via PCA on Local Neighborhoods ↗

For a dataset $X \in \mathbb{R}^{m \times n}$, pick a point $x \in X$.
For each $\varepsilon_i \in \{\varepsilon_1, \dots, \varepsilon_k\}$

1. Find points within a distance ε_i of x.
$$N\varepsilon_i = \{x_j \in X \mid d(x_j, x) \le \varepsilon_i\}$$
2. Compute the PCA Eigenvalues for ${N\varepsilon_i}^T N\varepsilon_i$ where $N\varepsilon_i = U\Sigma V^T$ (by SVD) and
$$\Sigma = \begin{pmatrix} \sqrt{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sqrt{\lambda_n} \end{pmatrix}$$
with eigenvalues $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_n$.

The **number of large eigenvalues** over a given neighborhood (or many) is the intrinsic dimensionality of the data over those neighborhoods.

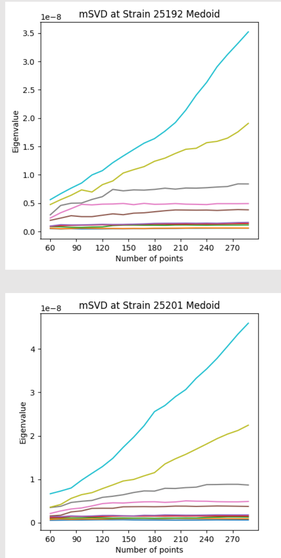


Figure 7: mSVD plots of two D. melanogaster wing strain medoids. There is a local, low-dimensional (2-3 dimensional) structure to our space.

Conclusions and Future Work

- We have shown that the 2-Wasserstein metric provides a novel and informative method for quantifying morphological variation.
- We aim to fully interpret the dimensions elucidated by mSVD.
- Future work is currently focused on generalizing this methodology to unequal positive measures using the Wasserstein-Fisher-Rao metric **[2,3]**.

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Acknowledgments

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